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## **1. Introduction**

This report presents the design of a complete network infrastructure for a medium-sized hospital environment. The hospital includes approximately 500 staff members and supports up to 1000 patients. As a result, the network must handle a large number of users, devices, and critical medical systems.

In modern healthcare environments, the network is no longer just a communication tool. It plays a direct role in patient care, especially with the use of medical imaging systems, operating room monitoring devices, and real-time data access. Therefore, the network must be highly reliable, secure, and capable of supporting high-bandwidth applications.

The goal of this design is to ensure efficient communication between different hospital departments such as administration, doctors, laboratories, and radiology, while also providing secure and high-speed connectivity to remote clinics through a robust WAN infrastructure.

In addition, the network design focuses on key requirements such as scalability, fault tolerance, and redundancy to eliminate single points of failure and ensure continuous operation. Modern technologies are applied selectively in critical areas to improve performance while maintaining a cost-effective solution.

## **2. Network Requirements**

### **a. Function Requirements**

The hospital network must provide a set of essential functions to support medical operations, ensure efficient communication, and enable integration between departments and remote clinics.

#### **1. Real-Time Data Transmission**

The network must support real-time transmission of critical medical data between departments such as operating rooms, radiology, and patient monitoring systems. This ensures that medical staff can access and update patient information instantly without delay.

#### **2. Communication Between Hospital Departments**

The network must enable seamless communication between all hospital departments, including administration, doctors, laboratories, and training units. This ensures efficient coordination and smooth workflow across the hospital.

#### **3. Connectivity with Remote Clinics**

The network must provide reliable connectivity between the main hospital and remote clinics. This allows remote medical staff to access centralized systems such as patient records, imaging data, and hospital services.

#### **4. Support for Medical Applications**

The network must support specialized medical applications such as radiology systems, patient monitoring devices, and hospital management systems. These systems require stable and continuous connectivity to function properly.

#### **5. High Bandwidth Support**

The network must handle large volumes of data traffic, especially departments such as radiology and laboratories that generate high data loads.

## **6. Inter-Department Data Sharing**

The network must allow efficient and secure data exchange between different departments. For example, doctors should be able to access lab results and imaging data when needed.

## **7. Centralized Access to Services**

The network must allow users to access centralized services such as servers, databases, and applications from different locations within the hospital.

## **8. End-User Device Connectivity**

The network must support connectivity for a wide range of end devices, including computers, laptops, printers, IP phones, and medical equipment.

## **9. Network Monitoring and Management**

The network must provide monitoring and management capabilities to track performance, detect faults, and ensure smooth operation of all network components.

## **b. Non-function requirement**

The non-functional requirements define the quality attributes and performance expectations of the hospital network. These requirements ensure that the system operates efficiently, securely, and reliably in a critical healthcare environment.

### **1. Security**

The network must provide a high level of security to protect sensitive patient data and prevent unauthorized access. It must ensure controlled access between departments, restrict unauthorized communication, and continuously monitor network traffic to detect and prevent potential threats. This is critical since the hospital handles highly confidential medical data.

### **1. Availability**

The network must ensure continuous operation with minimal downtime, as hospital services are required 24/7. It should maintain service availability even during failures and avoid any single point of failure that could disrupt the entire system.

### **2. Performance**

The network must support high-speed data transmission with low latency, especially for real-time applications such as medical imaging and patient monitoring. It should ensure efficient handling of network traffic and provide consistent performance under heavy load conditions.

### **4. Scalability**

The network must support future expansion in terms of users, devices, and services. It should allow easy integration of new components and accommodate growth without requiring major changes to the existing infrastructure.

## **5. Reliability**

The network must operate consistently without frequent failures. It should ensure stable performance and maintain dependable operation under different conditions.

## **6. Maintainability**

The network should be easy to manage, troubleshoot, and upgrade. It must support efficient monitoring and allow quick identification and resolution of issues.

## **7. Fault Tolerance**

The network must continue operating even in the event of failures. It should be able to handle faults gracefully and maintain service continuity without significant disruption.

## **8. Quality of Service (QoS)**

The network must prioritize critical applications over less important traffic. It should ensure that time-sensitive medical data receives higher priority and is delivered with minimal delay.

## **9. Disaster Recovery**

The network must include a disaster recovery mechanism to restore services quickly in case of major failures or system outages, ensuring minimal data loss and service disruption.

### 3. Network Design Overview

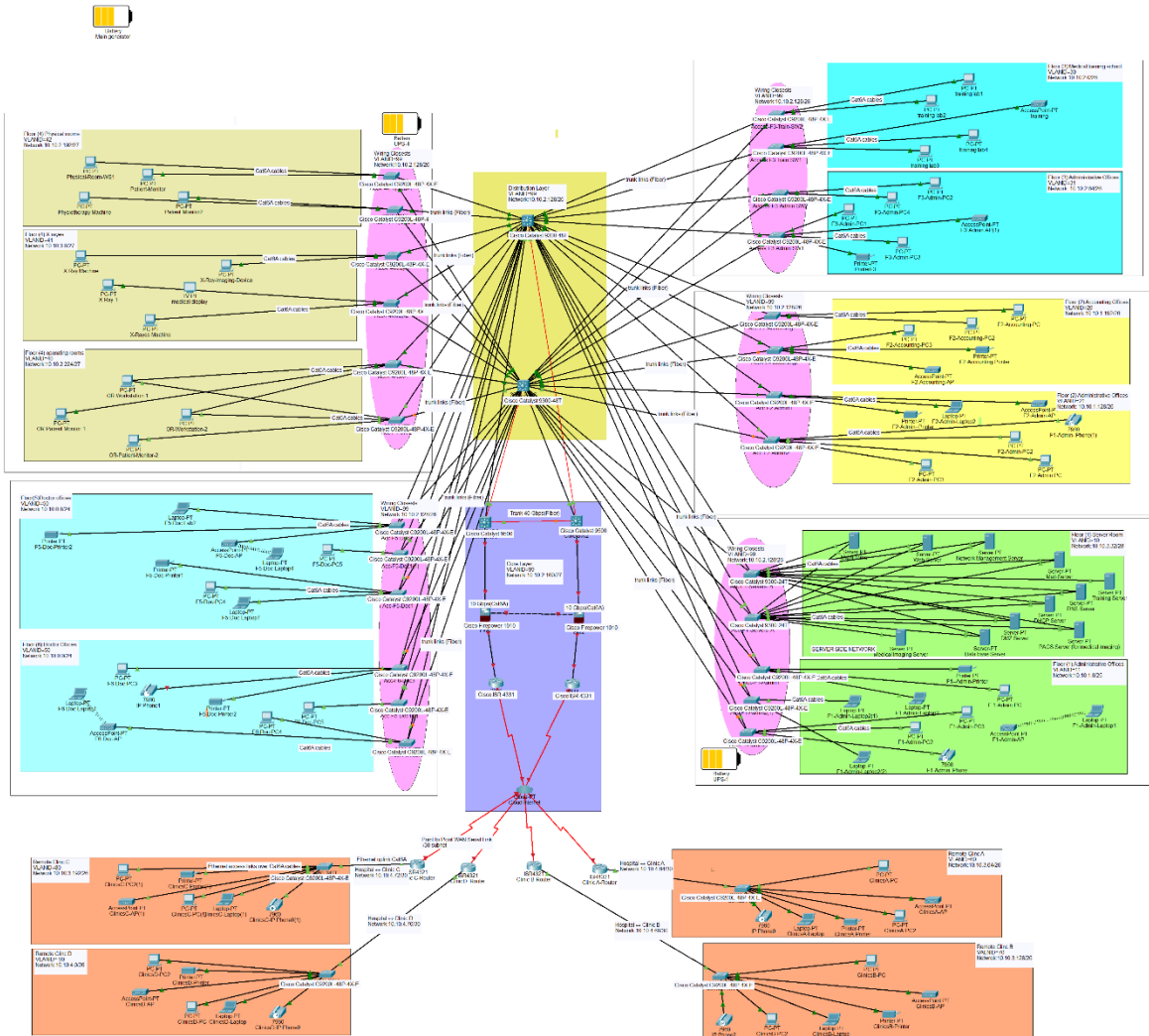


Figure 1: Hospital Network Design Overview

The proposed hospital network design follows a hierarchical architecture consisting of three main layers: core, distribution, and access. This structured approach is adopted to ensure scalability, high performance, and simplified network management across the entire hospital environment.

The core layer acts as the backbone of the network, providing high-speed connectivity between distribution switches, the centralized server room, and WAN connections to remote clinics. To enhance reliability and eliminate single points of failure, the core layer is designed using redundant core switches and high-speed fiber links, ensuring continuous and stable communication across the network.

The distribution layer is deployed per floor, where each floor contains its own distribution switches to aggregate traffic from access switches. This layer is responsible for inter-VLAN routing and policy enforcement, ensuring efficient communication between departments while maintaining control, segmentation, and optimized traffic flow.

The access layer connects all end-user devices throughout the hospital, including computers, medical equipment, printers, IP phones, and wireless access points. These devices are connected via Ethernet access links to provide reliable and structured connectivity within each department and floor.

The hospital network is logically segmented using VLANs to separate different departments such as administration, doctors' offices, laboratories, operating rooms, radiology, and training areas. This segmentation enhances both security and traffic management by isolating broadcast domains and reducing unnecessary network traffic.

A centralized server room is included in the design to host critical systems such as database servers, DNS, DHCP, web servers, and other application services. This ensures centralized management and efficient access to essential resources across the network.

In addition, the design incorporates WAN connectivity to link the main hospital with multiple remote clinics, enabling secure and efficient access to centralized systems and services from remote locations. This connectivity is implemented using MPLS-based WAN links, which provide dedicated, high-reliability communication between the hospital and remote clinics. MPLS is

selected due to its ability to deliver low latency, consistent performance, and Quality of Service (QoS), which are essential for real-time medical applications and large data transfers such as medical imaging.

To improve service continuity, the design includes centralized UPS support for critical hospital infrastructure, particularly in the core network area. In addition, dedicated UPS units are provided for the first floor and the fourth floor due to the critical nature of the services hosted there. The first floor contains the server farm and centralized resources, while the fourth floor includes operating and imaging-related areas that require continuous network availability. This improves availability, fault tolerance, and protection against sudden power loss.

This power backup strategy ensures that critical hospital services remain operational during short-term power interruptions and reduces the risk of service disruption in highly sensitive areas.

Finally, redundancy is implemented in both the core and distribution layers using dual links and redundant devices to improve fault tolerance and ensure high availability. Overall, this design provides a scalable, reliable, and high-performance network infrastructure suitable for a modern healthcare environment.

### 3. VLAN Design

*Table 1: VLAN Design*

| Floor   | VLAN ID | Network Name   | Purpose                                                                 | Number of Devices |
|---------|---------|----------------|-------------------------------------------------------------------------|-------------------|
| Floor 1 | VLAN 10 | Server Farm    | 7 Shared Servers + Mainframe + Database Server + Medical Imaging Server | 10 Servers        |
| Floor 1 | VLAN 11 | Admin Ground   | Administrative Offices Floor                                            | 120               |
| Floor 2 | VLAN 20 | Accounting     | Accounting Department                                                   | 50                |
| Floor 2 | VLAN 21 | Admin 2        | Administrative Offices Floor                                            | 60                |
| Floor 3 | VLAN 30 | Medical School | Medical Training School                                                 | 50                |
| Floor 3 | VLAN 31 | Admin 3        | Administrative Offices Floor                                            | 50                |

|            |         |            |                        |                      |
|------------|---------|------------|------------------------|----------------------|
| Floor 4    | VLAN 40 | Operating  | Operating              | 20                   |
| Floor 4    | VLAN 41 | X- Rays    | X- Rays                | 20                   |
| Floor 4    | VLAN 42 | physical   | physical               | 30                   |
| Floor 5 &6 | VLAN 50 | Doctors 5  | Doctors' Offices Floor | 200                  |
| Private    | VLAN 99 | Management | Network Management     | Network devices only |

## 4. IP Addressing & Subnetting

Table 2: IP Addressing & Subnetting

| Floor   | VLAN ID | #Hosts               | CIDR | Subnet Mask     | Network Addresses | Range of Usable IPs        | Broadcast   | Default Gateway |
|---------|---------|----------------------|------|-----------------|-------------------|----------------------------|-------------|-----------------|
| Floor 1 | 10      | servers=10           | /28  | 255.255.255.240 | 10.10.3.32/28     | 10.10.3.33 to 10.10.3.46   | 10.10.3.47  | 10.10.3.33      |
| Floor 1 | 11      | Admin device=120     | /25  | 255.255.255.128 | 10.10.1.0/25      | 10.10.1.1 to 10.10.1.126   | 10.10.1.127 | 10.10.1.1       |
| Floor 2 | 21      | Admin device2=60     | /26  | 255.255.255.192 | 10.10.1.128/26    | 10.10.1.129 to 10.10.1.190 | 10.10.1.191 | 10.10.1.129     |
| Floor 2 | 20      | Accounting device=50 | /26  | 255.255.255.192 | 10.10.1.192/26    | 10.10.1.193 to 10.10.1.254 | 10.10.1.255 | 10.10.1.193     |
| Floor 3 | 30      | Medical Training=50  | /26  | 255.255.255.192 | 10.10.2.0/26      | 10.10.2.1 to 10.10.2.62    | 10.10.2.63  | 10.10.2.1       |
| Floor 3 | 31      | Admin device3=50     | /26  | 255.255.255.192 | 10.10.2.64/26     | 10.10.2.65 to 10.10.2.126  | 10.10.2.127 | 10.10.2.65      |
| Floor 4 | 40      | Operating room=20    | /27  | 255.255.255.224 | 10.10.2.224/27    | 10.10.2.225 to 10.10.2.254 | 10.10.2.255 | 10.10.2.225     |
| Floor 4 | 41      | X- rays =20          | /27  | 255.255.255.224 | 10.10.3.0/27      | 10.10.3.1 to 10.10.3.30    | 10.10.3.31  | 10.10.3.1       |
| Floor 4 | 42      | Physical room=30     | /27  | 255.255.255.224 | 10.10.2.192/27    | 10.10.2.193 to 10.10.2.222 | 10.10.2.223 | 10.10.2.193     |

|           |    |                                                                          |     |                 |                |                            |             |             |
|-----------|----|--------------------------------------------------------------------------|-----|-----------------|----------------|----------------------------|-------------|-------------|
| Floor 5&6 | 50 | Doctor Offices=200                                                       | /24 | 255.255.255.0   | 10.10.0.0/24   | 10.10.0.1 to 10.10.0.254   | 10.10.0.255 | 10.10.0.1   |
| Private   | 99 | Management devices only (switches, core switches, router, firewall) = 40 | /26 | 255.255.255.192 | 10.10.2.128/26 | 10.10.2.129 to 10.10.2.190 | 10.10.2.191 | 10.10.2.129 |

Remote Clinics

Table 3: Remote Clinics

| Clinics  | VLAN ID | #Hosts | CIDR | Subnet Mask     | Network Addresses | Range of Usable IPs        | Broadcast   | Default Gateway |
|----------|---------|--------|------|-----------------|-------------------|----------------------------|-------------|-----------------|
| Clinic A | 60      | 30 -50 | /26  | 255.255.255.192 | 10.10.3.64/26     | 10.10.3.65 to 10.10.3.126  | 10.10.3.127 | 10.10.3.65      |
| Clinic B | 70      | 30-50  | /26  | 255.255.255.192 | 10.10.3.128/26    | 10.10.3.129 to 10.10.3.190 | 10.10.3.191 | 10.10.3.129     |
| Clinic C | 80      | 30-50  | /26  | 255.255.255.192 | 10.10.3.192/26    | 10.10.3.193 to 10.10.3.254 | 10.10.3.255 | 10.10.3.193     |
| Clinic D | 90      | 30-50  | /26  | 255.255.255.192 | 10.10.4.0/26      | 10.10.4.1 to 10.10.4.62    | 10.10.4.63  | 10.10.4.1       |

WAN Links

Table 4: WAN Links

| Link                | CIDR | Network Address | Usable IPs              | Broadcast  |
|---------------------|------|-----------------|-------------------------|------------|
| Hospital ↔ Clinic A | /30  | 10.10.4.64/30   | 10.10.4.65 – 10.10.4.66 | 10.10.4.67 |
| Hospital ↔ Clinic B | /30  | 10.10.4.68/30   | 10.10.4.69 – 10.10.4.70 | 10.10.4.71 |

|                     |     |               |                         |            |
|---------------------|-----|---------------|-------------------------|------------|
| Hospital ↔ Clinic C | /30 | 10.10.4.72/30 | 10.10.4.73 – 10.10.4.74 | 10.10.4.75 |
| Hospital ↔ Clinic D | /30 | 10.10.4.76/30 | 10.10.4.77 – 10.10.4.78 | 10.10.4.79 |

## 5. Physical Network Design

### Common Access Layer Design

All end-user devices across the hospital, including PCs, laptops, IP phones, printers, and medical equipment, are connected to access switches using Ethernet access links. Each device is assigned to a single VLAN to ensure proper traffic segmentation and organized communication within departments.

To improve performance and avoid congestion, devices within each VLAN are distributed across multiple access switches. This prevents overloading individual switches and ensures efficient traffic handling.

Each access switch is connected to the distribution layer using dual uplinks configured as Ethernet trunk links (IEEE 802.1Q). These trunk links carry multiple VLANs while maintaining logical separation between network segments.

The use of dual uplinks provides redundant communication paths. In case of link or device failure, traffic is automatically redirected through the alternative path, ensuring continuous operation.

Since multiple paths exist in the network, Spanning Tree Protocol (STP) is implemented to prevent switching loops. STP ensures that only one active path is used while keeping backup paths available for failover.

This design supports key network requirements, including high availability, fault tolerance, scalability, and efficient traffic management.

## Core Layer:

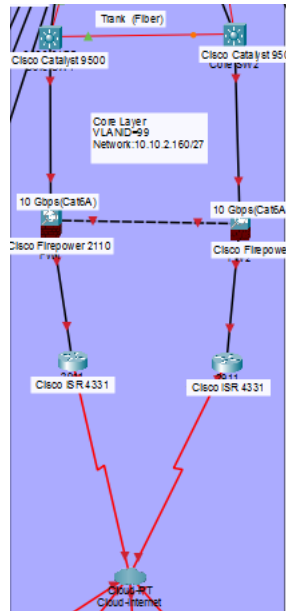


Figure 2: Core Layer

The core layer represents the backbone of the entire hospital network. It is responsible for forwarding traffic between distribution switches, the centralized server segment, and the WAN connections to remote clinics.

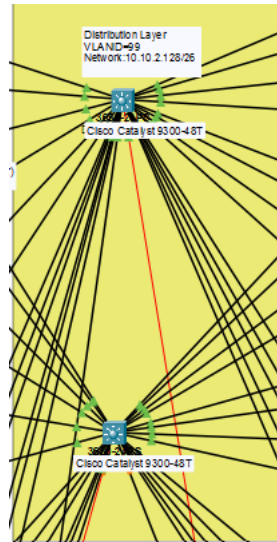
In this design, the core layer is implemented using redundant core switches to eliminate any single point of failure. These switches are interconnected using high-speed links to ensure fast data forwarding and minimal latency across the network.

The core layer is connected to the distribution layer using high-speed trunk links to handle aggregated traffic coming from multiple floors. These links provide sufficient bandwidth to prevent bottlenecks and ensure efficient data transmission.

For external connectivity, the core layer connects to the WAN through firewalls and edge routers. One firewall operates normally (active). The second firewall remains on standby and takes over if the primary fails.

Firewalls are critical for security and network access control. Redundancy ensures that failure of a firewall does not interrupt communication or compromise security.

## Distribution Layer:



*Figure 3: Distribution Layer*

The distribution layer acts as the intermediate layer between the access layer and the core layer. Its main function is to aggregate traffic from access switches and prepare it for forwarding to the core layer.

In this design, the distribution layer consists of two switches to ensure redundancy and eliminate single points of failure. Each access switch is connected to both distribution switches using dual uplinks configured as trunk links (IEEE 802.1Q).

These trunk links carry traffic from multiple VLANs while maintaining proper segmentation between departments. This ensures efficient traffic management and prevents unnecessary broadcast traffic.

The distribution switches are also interconnected using a high-speed link, allowing direct communication and ensuring continuous operation even if one path fails.

From the distribution layer, aggregated traffic is forwarded to the core layer using high-speed links, ensuring efficient data transmission without bottlenecks.

## First Floor Network:

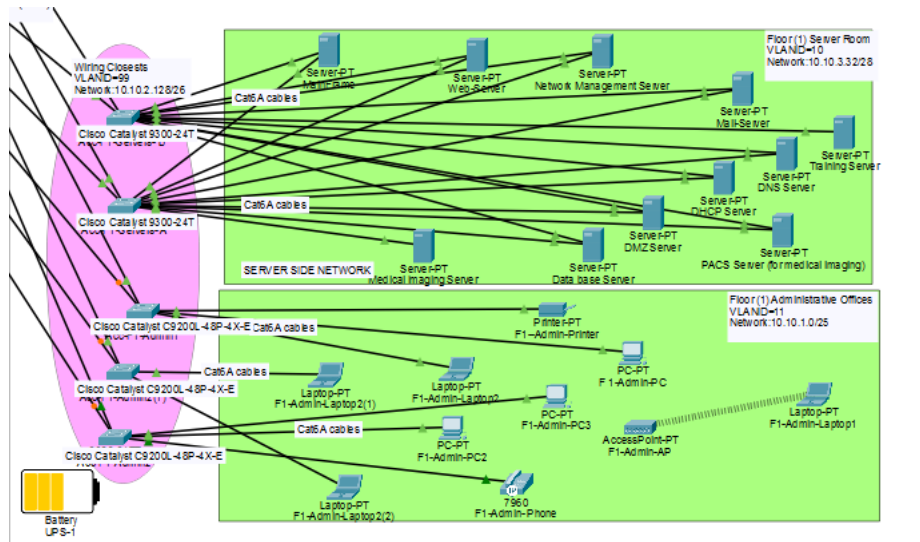


Figure 4: First Floor

The first floor is divided into two VLANs: VLAN 10 for the server farm and VLAN 11 for administrative offices. This separation isolates critical server traffic from user traffic, improving both security and performance.

The server farm includes critical systems such as database, DNS, DHCP, and medical imaging servers. These servers are connected using redundant links to ensure continuous availability, as they host essential hospital services.

Administrative users are distributed across multiple access switches to support efficient traffic handling and ensure uninterrupted operation.

Uninterruptible Power Supply (UPS) systems are used to provide backup power for critical devices, ensuring continuous operation during power failures.

## Second Floor Network:

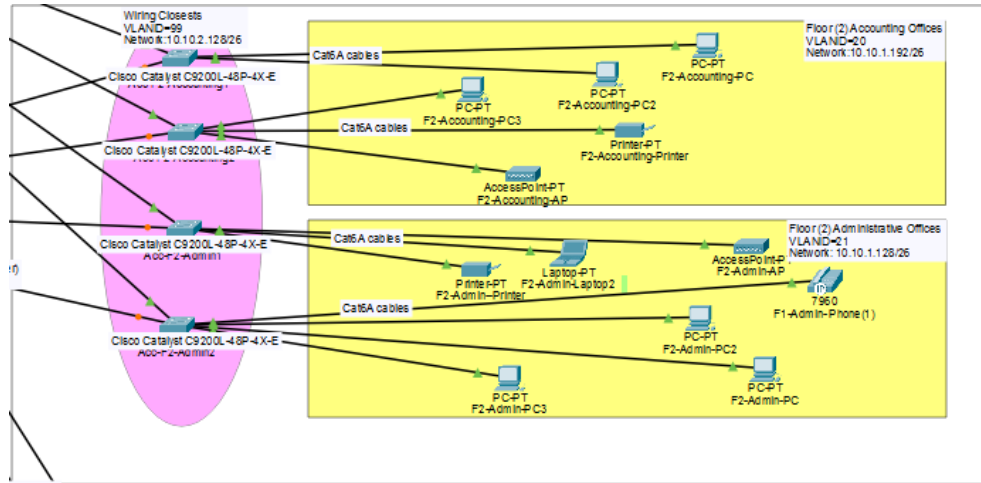


Figure 5: Second Floor

The second floor is divided into two VLANs: VLAN 20 for accounting and VLAN 21 for administrative offices. This separation ensures proper isolation between financial and administrative traffic, improving security and control.

Devices are distributed across multiple access switches to ensure efficient load handling and prevent congestion, especially in high-usage areas such as accounting.

## Third Floor Network:

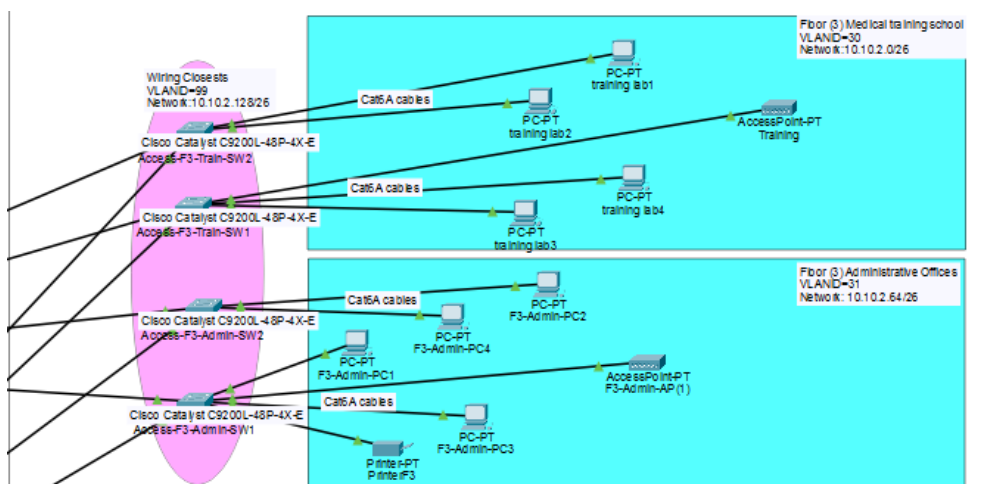


Figure 6: Third Floor

The third floor supports both medical training labs (VLAN 30) and administrative offices (VLAN 31). Training labs generate dynamic and high-volume traffic due to simultaneous user activity.

To handle this, devices are distributed across multiple switches to ensure smooth performance during peak usage and maintain system stability.

### Fourth Floor Network:

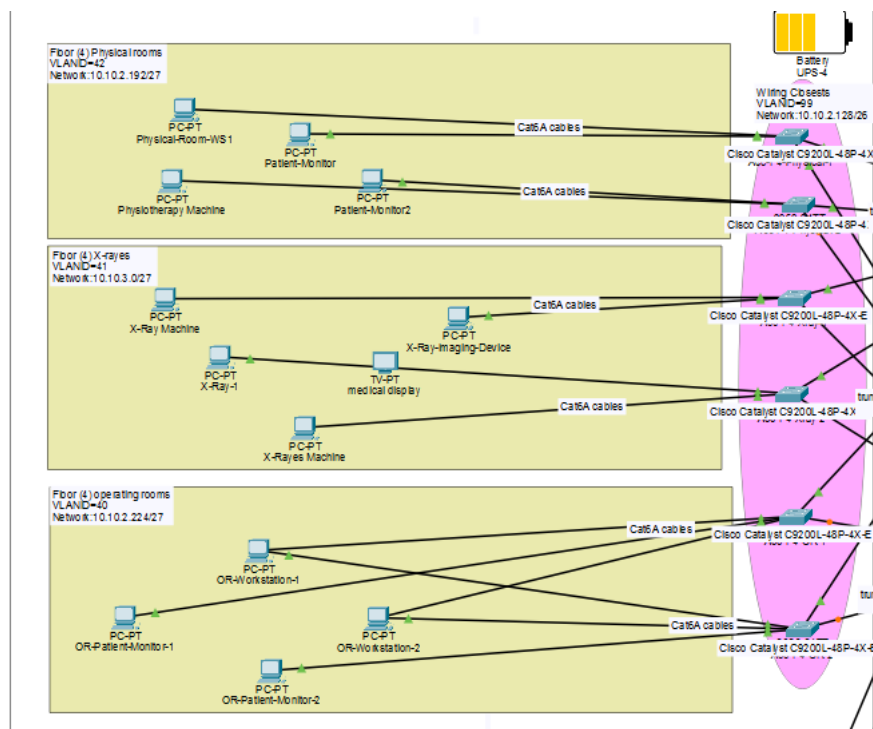


Figure 7: Fourth Floor

The fourth floor includes critical medical departments: operating rooms (VLAN 40), X-ray rooms (VLAN 41), and physical therapy (VLAN 42). These departments require reliable and continuous connectivity due to the critical nature of medical operations.

Key medical devices are connected using redundant links to ensure uninterrupted operation, especially in operating and imaging environments.

Uninterruptible Power Supply (UPS) systems are used to provide backup power for critical devices, ensuring continuous operation during power failures.

## Fifth & Sixth Floor Network:

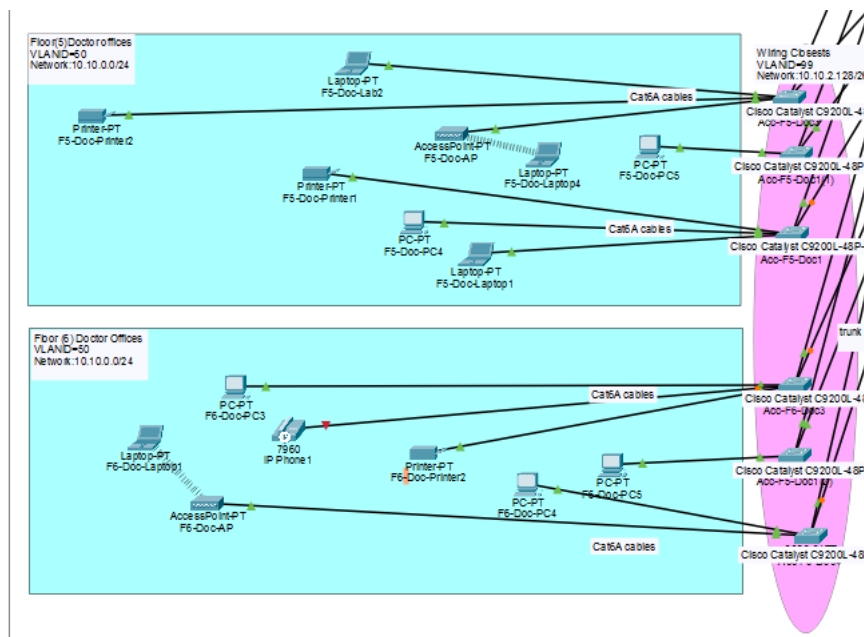


Figure 8: Fifth & Six Floor

The fifth and sixth floors are dedicated to doctors' offices and operate within a single VLAN (VLAN 50). This allows seamless communication between doctors across both floors.

Devices are distributed across multiple access switches to handle the large number of users and ensure scalability and performance.

## Remote Clinics:

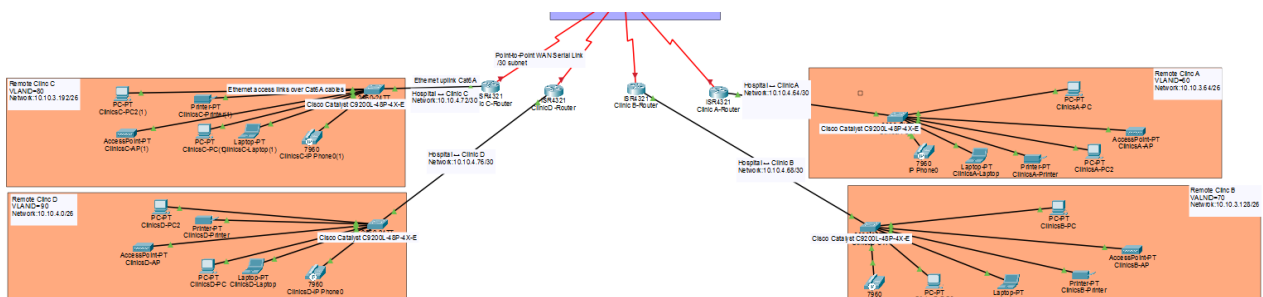


Figure 9: Remote clinics

Each remote clinic operates as an independent local network connected to the hospital via WAN links. Traffic flows from end devices to access switches, then to the local router, and finally to the hospital core.

Each clinic uses a simple structure to reduce complexity while maintaining reliable connectivity to centralized hospital services.

## 6. Network Devices & Equipment

*Table 5: Network Devices & Equipment*

| Layer                      | Device              | Model                          | Key Features                                                                                | Purpose in Design                                                                                                                       |
|----------------------------|---------------------|--------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core Layer</b>          | Core Switch         | Cisco Catalyst 9500            | Supports 10G/40G/100G uplinks, high throughput, Layer 3 routing, high switching capacity    | Acts as the backbone of the network, handling aggregated traffic from all distribution switches and ensuring high-speed data forwarding |
| <b>Core Layer</b>          | Firewall            | Cisco Firepower 2110           | High throughput, IPS, AVC, URL filtering, MPLS support, 8×1GE ports, active-standby support | Secures the network edge by inspecting and filtering all incoming and outgoing traffic and protecting sensitive medical data            |
| <b>Core Layer</b>          | WAN Router          | Cisco ISR 4331                 | Supports WAN services, routing protocols (OSPF, EIGRP, BGP), MPLS, modular interfaces       | Connects the hospital network to remote clinics and external networks with reliable and secure communication                            |
| <b>Distribution Layer</b>  | Distribution Switch | Cisco Catalyst 9300-48T        | 48×1GE ports, Layer 3 support, 10G uplinks, high performance                                | Aggregates traffic from access switches and performs inter-VLAN routing and traffic control                                             |
| <b>Access Layer</b>        | Access Switch       | Cisco Catalyst C9200L-48P-4X-E | 48×1GE ports, PoE support, 4×10G uplinks, high scalability                                  | Connects end-user devices such as PCs, printers, IP phones, and access points                                                           |
| <b>Server Access Layer</b> | Server Switch       | Cisco Catalyst 9300-24T        | 24×1GE ports, 10G uplinks, stacking support, high performance                               | Connects servers with high-speed and reliable links, supporting redundancy and efficient traffic handling                               |
| <b>WAN (Clinics)</b>       | Branch Router       | Cisco ISR 4321                 | WAN connectivity, MPLS support, cost-effective, branch design                               | Provides connectivity between each remote clinic and the hospital network                                                               |

|                       |              |                        |                                                                    |                                                                                      |
|-----------------------|--------------|------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Wireless Layer</b> | Access Point | Cisco Catalyst 9115AXI | Wi-Fi 6, supports high user density, PoE powered, high performance | Provides wireless connectivity for mobile devices across hospital floors and clinics |
|-----------------------|--------------|------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------|

# 7. Cabling Infrastructure

Table 6: Cabling Infrastructure

| Cable Type                                  | Usage / Connections                                                                                                                                                           | Key Features                                                                                               | Purpose in Design                                                                                                     |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Ethernet Copper (Cat6A)</b>              | <p>End devices → Access switches</p> <p>Wireless APs → Access switches</p> <p>IP phones &amp; printers → Access switches</p> <p>Access switches → Local routers (clinics)</p> | <p>Supports Gigabit Ethernet</p> <p>Low latency Cost-effective</p> <p>Easy installation</p>                | Used for short-distance connections within floors to provide reliable and efficient connectivity for end-user devices |
| <b>Fiber Optic Cable</b>                    | <p>Access → Distribution (between floors)</p> <p>Distribution → Core</p> <p>Core → Firewalls</p> <p>Firewalls → WAN routers</p>                                               | <p>Very high bandwidth</p> <p>Long-distance support</p> <p>Low signal loss Immune to interference</p>      | Used for backbone connections to carry large volumes of aggregated traffic across network layers                      |
| <b>WAN Serial Links</b>                     | Router → ISP (Cloud) connections (remote clinics)                                                                                                                             | <p>Reliable point-to-point communication</p> <p>Supports /30 addressing</p> <p>Stable WAN connectivity</p> | Used to connect remote clinic routers to the service provider network over long distances                             |
| <b>High-Speed Ethernet (Backbone Links)</b> | <p>Distribution ↔ Core</p> <p>Core ↔ Firewalls</p>                                                                                                                            | <p>High bandwidth (10G or more)</p> <p>Low latency Fast data transfer</p>                                  | Used for critical backbone communication to prevent bottlenecks and support high traffic loads                        |

## **8. Alternative Design**

### **Collapsed Core Architecture**

An alternative design for the proposed hospital network is the collapsed core architecture, in which the core and distribution layers are combined into a single layer. This approach simplifies the hierarchical structure by eliminating the dedicated distribution layer and assigning its functions to the core layer.

In this design, all access switches deployed across the six floors of the hospital are connected directly to a pair of high-performance multilayer core switches using high-speed uplinks. These core switches are responsible for performing traffic aggregation, inter-VLAN routing, and forwarding traffic toward internal resources such as servers, as well as external networks through the firewall and WAN infrastructure.

Communication in this architecture follows a direct path. Traffic generated by end-user devices flows from the devices to the access switches, and then directly to the collapsed core layer without passing through a separate distribution layer. This reduces the number of hops in the network, which can improve latency and simplify traffic flow.

From a resource perspective, the design includes two core switches operating in a redundant configuration to provide failover in case of device failure. Access switches are deployed on each floor to connect end-user devices, while critical servers such as database, DHCP, DNS, and medical imaging systems are connected through dedicated server access switches to ensure efficient and reliable access to centralized services.

In terms of cabling, Cat6A cables are used for connecting end-user devices to access switches due to their suitability for short-distance, high-speed communication within the floor environment. High-speed uplinks between access switches and the core layer are implemented using fiber optic connections to support large volumes of aggregated traffic and ensure high bandwidth and reliability.

For WAN connectivity, an alternative approach using VPN over the public internet can be considered instead of MPLS-based connections. While VPN provides a cost-effective and flexible solution, it may introduce higher latency and less predictable performance compared to dedicated WAN links, making it less suitable for real-time medical applications.

Although the collapsed core design reduces the number of devices, simplifies the network topology, and lowers implementation cost, it introduces certain limitations. Since all traffic aggregation and routing functions are handled by the core layer, this creates a potential performance bottleneck under high traffic conditions. In addition, scalability is reduced, as expanding the network would place additional load on the same layer.

Therefore, while the collapsed core architecture represents a valid and realistic alternative, it is less suitable for this project compared to the selected three-tier hierarchical design, which provides better scalability, improved fault isolation, and more efficient handling of high-volume and real-time medical traffic.

## 9. Budget

Table 7: Budget

| Category                       | Item                                      | Quantity | Unit Cost (KD) | Total Cost |
|--------------------------------|-------------------------------------------|----------|----------------|------------|
| <b>Core Switch</b>             | Cisco Catalyst 9500                       | 2        | 3,000          | 6,000 KD   |
| <b>Distribution Switch</b>     | Cisco Catalyst 9300-48T                   | 2        | 700            | 1,400 KD   |
| <b>Firewall</b>                | Cisco Firepower 2110                      | 2        | 1,200          | 2,400 KD   |
| <b>WAN Router</b>              | Cisco ISR 4331                            | 2        | 350            | 700 KD     |
| <b>Access Switches</b>         | Cisco Catalyst C9200L-48P                 | 23       | 500            | 11,500 KD  |
|                                | Cisco Catalyst 9300-24T                   | 2        | 550 KD         | 1,100 KD   |
| <b>Router (Remote Clinics)</b> | Cisco ISR 4321 Integrated Services Router | 4        | 400 KD         | 1,600 KD   |

|                     |                                     |          |        |          |
|---------------------|-------------------------------------|----------|--------|----------|
| <b>Access Point</b> | Cisco Catalyst 9115AXI Access Point | 11       | 300 KD | 3,300    |
| <b>Cable</b>        | Cat6A                               | 700 runs | 95 KD  | 9,000 KD |
|                     | Fiber Optic                         | 57 links | 28 KD  | 1,600 KD |
|                     | WAN Serial                          | 6 links  | 200 KD | 1,200 KD |

### Calculation Basis for Cabling Costs:

**Cat6A Cable:** Estimated using average 40 m per run and 305 m per Cat6A box

**Fiber Optic Cable:** Estimated based on the number of backbone/uplink fiber connections

**WAN Serial Links:** One point-to-point WAN link per remote clinic

### Servers Cost:

*Table 8: Servers Cost*

| Category       | Item                      | Quantity | Unit Cost (KD) | Total Cost |
|----------------|---------------------------|----------|----------------|------------|
| <b>Servers</b> | Database Server           | 1        | 2,500          | 2,500 KD   |
| <b>Servers</b> | Medical Imaging Server    | 1        | 3,000          | 3,000 KD   |
| <b>Servers</b> | Application / Web Server  | 1        | 2,000          | 2,000 KD   |
| <b>Servers</b> | Mail Server               | 1        | 1,500          | 1,500 KD   |
| <b>Servers</b> | DNS & DHCP Server         | 1        | 1,200          | 1,200 KD   |
| <b>Servers</b> | Network Management Server | 1        | 1,500          | 1,500 KD   |
| <b>Servers</b> | Training Server           | 1        | 1,500          | 1,500 KD   |

**Total Cost: 53,000KD**

## 10.2 Services Budget

Table 9: Services Budget

| Category                                 | Description                    | Cost (KD) | Justification                                                                                                                                                                                    |
|------------------------------------------|--------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hospital Information System (HIS)</b> | Hospital management system     | 60,000    | HIS solutions are typically enterprise-level and quote-based with no fixed pricing. The cost depends on users and integrations, so a mid-range estimate was selected for a medium-sized hospital |
| <b>PACS (Medical Imaging System)</b>     | Imaging storage and management | 45,000    | PACS systems are resource-intensive due to high-resolution medical imaging storage, increasing cost, but scaled to a medium hospital size                                                        |
| <b>Learning Management System (LMS)</b>  | Educational platform           | 4,000     | A lower cost is selected since open-source platforms like Moodle reduce licensing costs, with expenses mainly in deployment and support                                                          |
| <b>Patient Portal</b>                    | Patient web system             | 8,000     | Moderate cost based on standard web functionality rather than complexity, despite large user base                                                                                                |
| <b>Network Security Services</b>         | Security services              | 18,000    | Cost increased due to sensitive healthcare data, but kept within mid-range rather than full enterprise deployment                                                                                |
| <b>Installation &amp; Cabling Labor</b>  | Installation and cabling       | 25,000    | Based on building size and number of endpoints, increasing labor requirements                                                                                                                    |

|                                               |                              |               |                                                                                                |
|-----------------------------------------------|------------------------------|---------------|------------------------------------------------------------------------------------------------|
| <b>Network Configuration &amp; Validation</b> | Setup and testing            | 15,000        | Reflects network complexity including VLANs, routing, and redundancy                           |
| <b>Network Monitoring</b>                     | Monitoring tools             | 8,000         | Mid-range cost based on entry-level pricing models suitable for medium networks                |
| <b>WAN Connectivity</b>                       | Remote connectivity          | 24,000 / year | WAN services are subscription-based and depend on bandwidth and number of remote sites         |
| <b>Cloud &amp; Backup Services</b>            | Backup and disaster recovery | 14,000 / year | Cloud services follow a pay-as-you-go model, with cost based on data size and backup frequency |
| <b>IT Support &amp; Maintenance</b>           | Maintenance and support      | 22,000 / year | Annual cost reflecting continuous operation, updates, and troubleshooting                      |
| <b>Database Management System</b>             | Database systems             | 40,000        | Higher cost due to large-scale structured and unstructured healthcare data                     |

The total one-time setup cost is approximately **223,000 KD**, while the annual operational cost is estimated at **60,000 KD per year**.

## **Conclusion**

In conclusion, the proposed network design successfully fulfills the key objectives of the project by providing a reliable, secure, and scalable infrastructure suitable for a medium-sized hospital environment. The design ensures sufficient bandwidth to support efficient data transfer between the main hospital and the remote clinics, particularly for critical healthcare applications such as real-time medical imaging and patient data management.

The overall network performance is enhanced using VLAN-based segmentation and a hierarchical architecture consisting of core, distribution, and access layers. This structured approach improves traffic management, reduces congestion, and allows for efficient communication between different departments within the hospital.

Scalability has been carefully considered, enabling the network to accommodate future expansion in terms of additional users, devices, and services without significantly affecting performance. This is particularly important given the expected growth in healthcare technologies and data requirements.

Furthermore, security requirements have been effectively addressed through logical segmentation, access control mechanisms, and the implementation of appropriate network protection measures. Redundancy has also been incorporated into the design to eliminate single points of failure, ensuring high availability and continuous operation of critical services.